BEHAVIOR-DRIVEN KNOWLEDGE GAP DETECTION USING GENERATIVE AI

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PROBLEM STATEMENT

Current learning systems evaluate progress using scores and completion, not actual understanding.

- Learners:
- think they understand when they don’t
- struggle silently
- follow fixed learning paths

This solution will not target individuals who drop topics from lact of interest that is more of an algorithmic approach that is affected by clickrate and how the student base is inherently brought to the course itself we target the group that drops a course due to reasons like difficulty to grasp concepts and to those that dont feel value addition.

Like Students, self learners, and professionals fail to build real conceptual clarity despite consuming content.

MOTIVATION

Today's learning platforms mostly rely on scores and quizzes to measure progress.

- But these don't always reflect real understanding.
- Many learners struggle without getting the right kind of support at the right time.
- There is a need for systems that can respond to how a learner is progressing and guide them accordingly

Generative AI makes this possible by:

- generating personalized practice problems
- producing alternate explanations
- adapting learning paths based on learner behavior

This helps move learning from fixed content → adaptive learning support

APPLICATION

This can work as a layer on top of existing learning platforms.

For example:

- A student is learning coding or math
- The system notices repeated retries, pauses, or hesitation
- It identifies where the learner is struggling
- It adjusts the next explanation or practice problem accordingly

Target users:

Students, online learners, coding beginners, exam prep learners.

PROPOSED SOLUTION

The idea is to understand learning through behavior, not just test results.

We look at things like:

- time spent on topics
- retry patterns
- hesitation while solving
- switching between concepts

Using this, the system:

- estimates what the learner understands
- identifies weak areas
- suggests the next best step to learn

HOW AND WHERE GENAI IS USED?

Generative AI is used inside the system to create adaptive learning support based on how a learner interacts with content.

Where it is used:

- Generating practice questions tailored to learner progress
- Creating alternate explanations for difficult concepts
- Producing the next recommended learning step
- Personalizing content difficulty and learning paths

How it works:

- The system observes learner interaction patterns
- GenAI generates targeted learning material and guidance
- The learning flow adapts continuously based on progress

Here, GenAI is used to produce learning interventions based on performance analysis data from behavioral models.

DATASETS

Data will be collected from learner interaction logs, problem-solving attempts, time spent on concepts, retry and error patterns, simulated learning behavior, and plagiarism indicators to train the system to generate adaptive learning support based on real learning patterns.

Datasets:

- EdNet (Student Interaction Logs):https://github.com/riiid/ednet
- Open University Learning Analytics Dataset (OULAD): https://analyse.kmi.open.ac.uk/open_dataset
- Student Engagement Dataset (SED):
- https://www.researchgate.net/publication/388136791_Student_Engagement_Dataset_SED_An_Online_Learning_Activity_Dataset
- ASSISTments Dataset (Educational Data Mining Dataset): https://sites.google.com/site/assistmentsdata/

EXPERIMENTS

We plan to test:

- effectiveness of generated practice questions in improving understanding
- improvement in learner performance after adaptive suggestions are generated
- reduction in repeated mistakes after receiving generated learning support
- learning speed compared to fixed learning paths
- quality and relevance of generated explanations and next learning steps
- learner engagement after personalized content is generated

Main focus: does the learner actually improve through generated learning support?

NOVELTY AND SCOPE TO SCALE

What makes this different:

- focuses on understanding how people learn, not just what they answer
- detects hidden gaps early
- adapts learning paths automatically

Where it can scale:

- EdTech platforms
- coding platforms
- exam prep tools
- skill learning apps
- corporate training systems

This can work anywhere learning happens.

THANK YOU